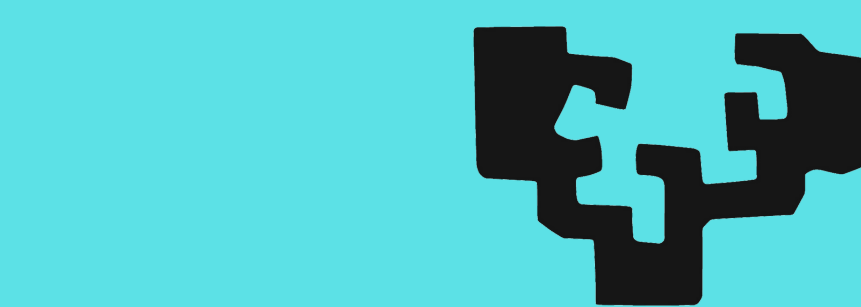




Eye-tracking the Experienter Thematic Role

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Introduction

Experienter thematic role designates the participant of an event that experiences the mental state denoted by the verb [1].

Dowty's [3] Proto-Roles theory: **Experienter** role fits in **Proto-Agent** category due to **sentience** entailment; it is less prototypical than Agent because Experienter only possesses one entailment.

Psycholinguistic evidence shows that Agents have different processing correlates than Themes [7]. Experienter processing has not been investigated yet.

Has the Experienter thematic role a specific processing pattern?

In order to determine this, we investigated argument structures with Experienter role [5,6]:

Psych verbs

Experienter-Theme
i.e. *love*

Agent-Experienter
i.e. *frighten*

Perceptual verbs

Experienter-Theme
i.e. *observe*

Psycholinguistic research suggests that psych verbs have different argument structures: *frighten* verbs involve causation whereas *love* verbs do not [4]. There is also evidence that [Agent-Experienter] structures involve larger processing cost than [Experienter-Theme] structures [2].

Methods

Eye-tracking

48 Spanish native speakers
(M=21 years).

Reading task with some comprehension questions.

20 verbs per condition controlled by length and frequency.

40 experimental items and 80 fillers. Acceptability task was made to control naturalness of experimental sentences.

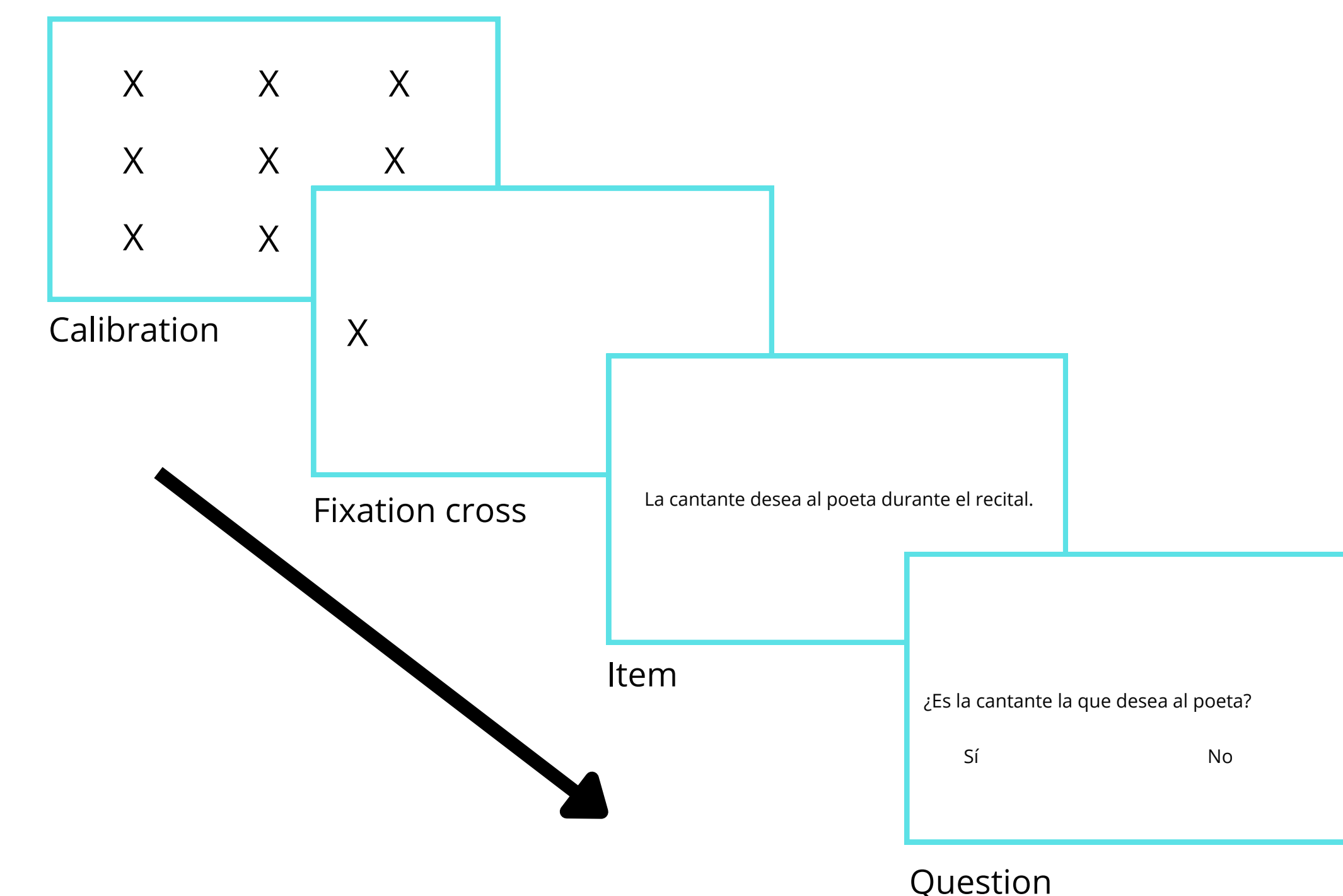
2x2 repeated measures design:

- **Verb Type** (Psych vs. Non-psych)
- **Argument Structure** (**Experienter-Theme** vs. **Agent-Theme/Experienter**)

Materials

Verb Type	Argument Structure	Subject Region	Verb Region	Object Region	Post-object Region	Last-word Region
Psych	Experienter-Theme	<i>La cantante</i>	<i>desea</i>	<i>al poeta</i>	durante el recital de	poesía.
Psych	Agent-Experienter	<i>La cantante</i>	<i>enamora</i>	<i>al poeta</i>	durante el recital de	poesía.
Non-psych	Experienter-Theme	<i>La cantante</i>	<i>contempla</i>	<i>al poeta</i>	durante el recital de	poesía.
Non-psych	Agent-Theme	<i>La cantante</i>	<i>abandona</i>	<i>al poeta</i>	durante el recital de	poesía.

The singer /desires/contemplates/makes fall in love/abandons /the poet/during the poetry/recital.



Results

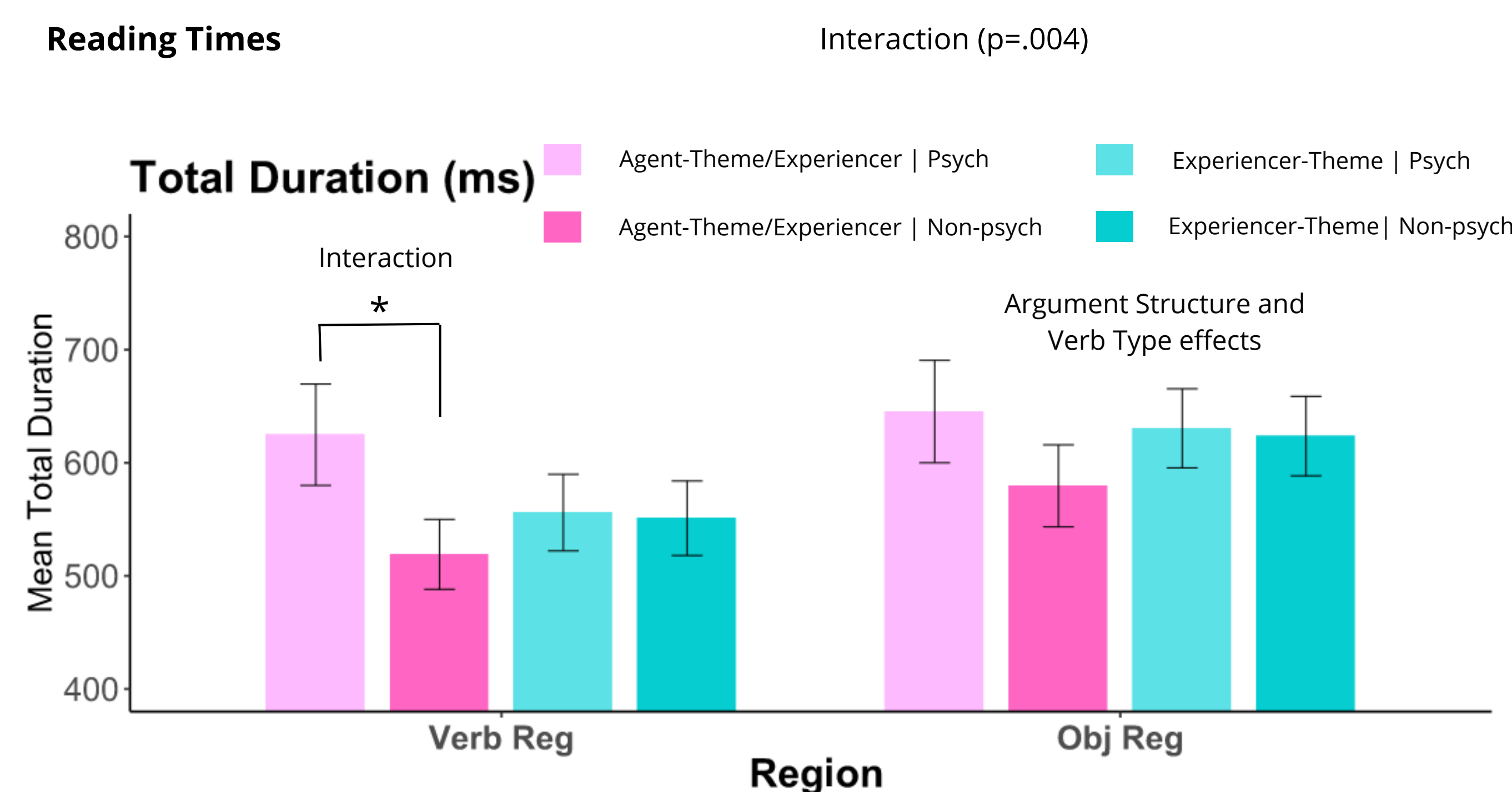
Remove fixations <80 ms and >800 ms.

Mixed models → LogGaze_Duration ~ verb_type * argument_structure + (1 | Participant) + (1 | Item)

AIC to choose the best fit model.

	Regions	Subject Region	Verb Region	Object Region	Post-object Region
Eye-tracking measures	Gaze Duration	X	Verb Type (p=.032)	Arg structure (p=.056)	X
	Regression Path Duration	X	Verb Type (p=.045)	Verb Type (p=.018)	X
	Re-reading Duration	X	Interaction (p=.008)	X	X
	Total Duration	X	Interaction (p=.032)	Verb Type (p=.032) Arg Structure (p=.034)	Arg Structure (p=.024)

Reading Times



Early measures (Gaze Duration and Regression Path Duration) - main effect of verb type in verb region: participants made larger fixation times on psych verbs than on non-psych verbs.

Late measures (Re-reading and Total Duration) - interaction in verb region: verb effect only appears in argument structures with Agent subject, not with Experienter subject. Participants made larger fixation times on Agent-Experienter structures than on Agent-Theme.

Argument structure effect in object and post-object regions: larger fixation times on sentences with Experienter-Theme argument structure than on those with Agent-Theme/Experienter.

Discussion and Conclusion

Experienter role involves increased processing load. It is not a prototypical Agent [3].

Experienter-Theme structures show larger processing cost than argument structures with Agent (fig. 1).

Agent-Experienter structures have two arguments that are candidates to be Proto-Agent, increasing difficulty to process compared to prototypical Agent-Theme structures (fig. 2).

We conclude that **Experienter role has specific processing correlates modulated by the other participant of the argument structure.**

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Acknowledgments

The Spanish Government (PID2021-124056NB-I00), the Basque Government (IT1439-22) and the University of the Basque Country (PIF21/263)



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